

Generalized Kanamori Hamiltonian is written as:

$$H_{GK} = U \sum_m n_{m\uparrow} n_{m\downarrow} + U' \sum_{m \neq m'} n_{m\uparrow} n_{m'\downarrow} + (U' - J) \sum_{m < m', \sigma} n_{m\sigma} n_{m'\sigma} \\ - J_x \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} + J_p \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow}^\dagger d_{m'\downarrow} d_{m'\uparrow}$$

One defines the total charge, spin, and orbital isospin generators:

$$N = \sum_{m\sigma} n_{m\sigma}, \quad \vec{S} = \frac{1}{2} \sum_m \sum_{\sigma\sigma'} d_{m\sigma}^\dagger \vec{\tau}_{\sigma\sigma'} d_{m\sigma'}, \quad L_m = i \sum_{m' m''} \sum_{\sigma} \epsilon_{m m' m''} d_{m'\sigma}^\dagger d_{m''\sigma} d_{m\sigma}$$

where $\vec{\tau}_{\sigma\sigma'}$ are the Pauli spin matrices. The operators in Σ are defined as:

$$\mathcal{D} = \sum_m n_{m\uparrow} n_{m\downarrow}, \quad X = \sum_{m \neq m'} n_{m\uparrow} n_{m'\downarrow}, \quad Y = \sum_{m < m', \sigma} n_{m\sigma} n_{m'\sigma},$$

$$F = \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow}, \quad P = \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow}^\dagger d_{m'\downarrow} d_{m'\uparrow}$$

Since the total number of particles is $N = \sum_{m\sigma} n_{m\sigma}$, one can guess $N^2 = \sum_{m\sigma} \sum_{m'\sigma'} n_{m\sigma} n_{m'\sigma'}$ can express $U \sum_m n_{m\uparrow} n_{m\downarrow}$. One can also

separate N^2 into two cases: (i) $(m, \sigma) = (m', \sigma')$ and (ii) $(m, \sigma) \neq (m', \sigma')$

Using $\sum_{\sigma} n_{m\sigma} = n_{m\uparrow} + n_{m\downarrow}$, one can write N^2 more explicitly:

$$N^2 = \left[\sum_m (n_{m\uparrow} + n_{m\downarrow}) \right] \left[\sum_{m'} (n_{m'\uparrow} + n_{m'\downarrow}) \right] \\ = \sum_m (n_{m\uparrow} + n_{m\downarrow})^2 + \sum_{m \neq m'} (n_{m\uparrow} + n_{m\downarrow}) (n_{m'\uparrow} + n_{m'\downarrow})$$

$$\textcircled{a} (n_{m\uparrow} + n_{m\downarrow})^2 = n_{m\uparrow}^2 + n_{m\uparrow} n_{m\downarrow} + n_{m\downarrow} n_{m\uparrow} + n_{m\downarrow}^2 \\ = n_{m\uparrow} + n_{m\downarrow} + 2 n_{m\uparrow} n_{m\downarrow} \quad (\text{using } n_{m\sigma}^2 = n_{m\sigma}, [n_{m\sigma}, n_{m\sigma'}] = 0)$$

If one subtracts $N = (n_{m\uparrow} + n_{m\downarrow})$ for both sides, one obtains 2d.

$$\textcircled{a} (n_{m\uparrow} + n_{m\downarrow}) (n_{m'\uparrow} + n_{m'\downarrow}) = \boxed{n_{m\uparrow} n_{m'\uparrow}} + \boxed{n_{m\uparrow} n_{m'\downarrow}} + \boxed{n_{m\downarrow} n_{m'\uparrow}} + \boxed{n_{m\downarrow} n_{m'\downarrow}}$$

Red terms: If one changes $m \leftrightarrow m'$, these terms are the same as X .

Blue terms: These terms are Y .

Therefore, N^2 can be represented with respect to X , Y , and $\mathcal{D}$.

$$\boxed{N^2 = N + 2(\mathcal{D} + X + Y)}$$

One can also guess that $\vec{S}$ can give an expression of F .
 The form of $\vec{S} = \frac{1}{2} \sum_m \sum_{\sigma\sigma'} d_{m\sigma}^\dagger \vec{\sigma}_{\sigma\sigma'} d_{m\sigma'}$ can be more explicitly written:

$$S^z = \frac{1}{2} \sum_m \sum_{\sigma\sigma'} d_{m\sigma}^\dagger \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} d_{m\sigma'} = \frac{1}{2} \sum_m (d_{m\uparrow}^\dagger d_{m\uparrow} - d_{m\downarrow}^\dagger d_{m\downarrow})$$

$$= \frac{1}{2} \sum_m (n_{m\uparrow} - n_{m\downarrow})$$

$$S^x = \frac{1}{2} \sum_m \sum_{\sigma\sigma'} d_{m\sigma}^\dagger \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} d_{m\sigma'} = \frac{1}{2} \sum_m (d_{m\uparrow}^\dagger d_{m\downarrow} + d_{m\downarrow}^\dagger d_{m\uparrow})$$

$$S^y = \frac{1}{2} \sum_m \sum_{\sigma\sigma'} d_{m\sigma}^\dagger \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} d_{m\sigma'} = \frac{i}{2} \sum_m (d_{m\uparrow}^\dagger d_{m\downarrow} - d_{m\downarrow}^\dagger d_{m\uparrow})$$

For each orbital degree of freedom, one can write S_m as $\vec{S} = \sum_m \vec{S}_m$. In addition, the square of S_m ($S_m^2 = S_m^{x^2} + S_m^{y^2} + S_m^{z^2}$) can be obtained by:

$$S_m^{x^2} = \frac{1}{4} (\cancel{d_{m\uparrow}^\dagger d_{m\downarrow} d_{m\uparrow}^\dagger d_{m\downarrow}} + d_{m\uparrow}^\dagger d_{m\downarrow} d_{m\downarrow}^\dagger d_{m\uparrow} + d_{m\downarrow}^\dagger d_{m\uparrow} d_{m\uparrow}^\dagger d_{m\downarrow} + \cancel{d_{m\downarrow}^\dagger d_{m\uparrow} d_{m\downarrow}^\dagger d_{m\uparrow}})$$

$$S_m^{y^2} = -\frac{1}{4} (\cancel{d_{m\uparrow}^\dagger d_{m\downarrow} d_{m\uparrow}^\dagger d_{m\downarrow}} - d_{m\uparrow}^\dagger d_{m\downarrow} d_{m\downarrow}^\dagger d_{m\uparrow} - d_{m\downarrow}^\dagger d_{m\uparrow} d_{m\uparrow}^\dagger d_{m\downarrow} + \cancel{d_{m\downarrow}^\dagger d_{m\uparrow} d_{m\downarrow}^\dagger d_{m\uparrow}})$$

$$S_m^{z^2} = \frac{1}{4} (n_{m\uparrow} - n_{m\uparrow} n_{m\downarrow} - n_{m\downarrow} n_{m\uparrow} + n_{m\downarrow}) = \frac{1}{4} (n_{m\uparrow} + n_{m\downarrow} - 2 n_{m\uparrow} n_{m\downarrow})$$

$$\therefore S_m^2 = \frac{1}{2} (d_{m\uparrow}^\dagger d_{m\downarrow} d_{m\downarrow}^\dagger d_{m\uparrow} + d_{m\downarrow}^\dagger d_{m\uparrow} d_{m\uparrow}^\dagger d_{m\downarrow})$$

$$+ \frac{1}{4} (n_{m\uparrow} + n_{m\downarrow} - 2 n_{m\uparrow} n_{m\downarrow})$$

$$= \frac{3}{4} (n_{m\uparrow} + n_{m\downarrow}) - \frac{3}{2} (n_{m\uparrow} n_{m\downarrow}) \quad (\text{using } \{d_{m\sigma}, d_{m\sigma}^\dagger\} = 1)$$

For different orbital m and m' ,

$$S_m^z S_{m'}^z = \frac{1}{4} (n_{m\uparrow} - n_{m\downarrow}) (n_{m'\uparrow} - n_{m'\downarrow})$$

$$= \frac{1}{4} (n_{m\uparrow} n_{m'\uparrow} + n_{m\downarrow} n_{m'\downarrow} - n_{m\uparrow} n_{m'\downarrow} - n_{m\downarrow} n_{m'\uparrow})$$

$$S_m^x S_{m'}^x = \frac{1}{4} (d_{m\uparrow}^\dagger d_{m\downarrow} + d_{m\downarrow}^\dagger d_{m\uparrow}) (d_{m'\uparrow}^\dagger d_{m'\downarrow} + d_{m'\downarrow}^\dagger d_{m'\uparrow})$$

$$= \frac{1}{4} (\cancel{d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\uparrow}^\dagger d_{m'\downarrow}} + d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} + d_{m\downarrow}^\dagger d_{m\uparrow} d_{m'\uparrow}^\dagger d_{m'\downarrow} + \cancel{d_{m\downarrow}^\dagger d_{m\uparrow} d_{m'\downarrow}^\dagger d_{m'\uparrow}})$$

$$S_m^\dagger S_{m'} = -\frac{1}{4} (d_{m\uparrow}^\dagger d_{m\downarrow} - d_{m\downarrow}^\dagger d_{m\uparrow}) (d_{m'\uparrow}^\dagger d_{m'\downarrow} - d_{m'\downarrow}^\dagger d_{m'\uparrow})$$

$$= -\frac{1}{4} (\cancel{d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\uparrow}^\dagger d_{m'\downarrow}} - d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} - d_{m\downarrow}^\dagger d_{m\uparrow} d_{m'\uparrow}^\dagger d_{m'\downarrow} + \cancel{d_{m\downarrow}^\dagger d_{m\uparrow} d_{m'\downarrow}^\dagger d_{m'\uparrow}})$$

$$\therefore S_m \cdot S_{m'} = \frac{1}{4} \left\{ n_{m\uparrow} n_{m'\uparrow} + n_{m\downarrow} n_{m'\downarrow} - n_{m\uparrow} n_{m'\downarrow} - n_{m\downarrow} n_{m'\uparrow} + 2 (d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} + d_{m\downarrow}^\dagger d_{m\uparrow} d_{m'\uparrow}^\dagger d_{m'\downarrow}) \right\}$$

Since

$$X = \sum_{m \neq m'} n_{m\uparrow} n_{m'\downarrow}, \quad Y = \sum_{m \leq m', \sigma} n_{m\sigma} n_{m'\sigma},$$

$$F = \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow},$$

The summation over $\sum_{m \neq m'}$ gives $2Y - 2X + 4F$ in the parenthesis, where one uses the Hermiticity of X and F , and $\sum_{m \leq m'} = 2 \sum_{m \neq m'}$. Including the S_m^2 , one obtains:

$$S^2 = \frac{3}{4} N - \frac{3}{2} D + \frac{1}{2} Y - \frac{1}{2} X + F$$

In order to express P , one expects to use N , S , and L (all of them).

First of all, $\varepsilon_{mn'm''}$ - Levi-Civita symbol satisfies the following relation:

$$\varepsilon_{abc} \varepsilon_{ade} = \delta_{bd} \delta_{ce} - \delta_{be} \delta_{cd}$$

Using this relation, one gets the expression of L^2 :

$$\begin{aligned} L^2 &= - \left(\sum_{m \hat{m}''} \sum_{\sigma} \varepsilon_{mn'm''} d_{m'\sigma}^\dagger d_{m''\sigma} \right) \left(\sum_{n \hat{n}'} \sum_{\sigma'} \varepsilon_{mn'n'} d_{n\sigma'}^\dagger d_{n'\sigma'} \right) \\ &= - \sum_{m' \hat{m}''} \sum_{n \hat{n}'} \sum_{\sigma \sigma'} (\delta_{m'n} \delta_{m''n'} - \delta_{m'n'} \delta_{m''n}) d_{m'\sigma}^\dagger d_{m''\sigma} d_{n\sigma'}^\dagger d_{n'\sigma'} \end{aligned}$$

$$= - \sum_{m'm''} \sum_{\sigma\sigma'} (d_{m'\sigma}^\dagger d_{m''\sigma} d_{m'\sigma'}^\dagger d_{m''\sigma'} - d_{m'\sigma}^\dagger d_{m''\sigma} d_{m''\sigma'}^\dagger d_{m'\sigma'})$$

using $\{d_{m\sigma}, d_{m'\sigma'}^\dagger\} = \delta_{mm'} \delta_{\sigma\sigma'}$, one obtains:

$$\begin{aligned} d_{m'\sigma}^\dagger d_{m''\sigma} d_{m'\sigma'}^\dagger d_{m''\sigma'} &= d_{m'\sigma}^\dagger (\delta_{m'm''} \delta_{\sigma\sigma'} - d_{m'\sigma}^\dagger d_{m''\sigma}) d_{m''\sigma'} \\ &= \delta_{m'm''} \delta_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m''\sigma'} - \text{red term} \\ &= + d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma} d_{m''\sigma'} \\ d_{m'\sigma}^\dagger d_{m''\sigma} d_{m''\sigma'}^\dagger d_{m'\sigma'} &= d_{m'\sigma}^\dagger (\delta_{\sigma\sigma'} - d_{m''\sigma}^\dagger d_{m''\sigma'}) d_{m'\sigma'} \\ &= \delta_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m'\sigma'} - \text{blue term} \end{aligned}$$

The red term survives only when $\sigma \neq \sigma'$. By changing $d_{m''\sigma} d_{m''\sigma'} \rightarrow -d_{m''\sigma'} d_{m''\sigma}$, one obtains $-d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma'} d_{m''\sigma}$. The summation over $\sum_{m'm''} \sum_{\sigma\sigma'}$ does not have any constraint. Thus, one must extract the contribution from $m'=m''$: $d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m'\sigma'} d_{m'\sigma}$ which is $n_{m'\sigma} n_{m'\sigma'}$. Additionally, it has the form of P with double counting from spin degree of freedom ($\sum_{\sigma\sigma'} = 2 \sum_{\sigma}$). In consequence,

$$\begin{aligned} \sum_{m'm''} \sum_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma'} d_{m''\sigma} &= \sum_{m'} \sum_{\sigma\sigma'} n_{m'\sigma} n_{m'\sigma'} + \sum_{m' \neq m''} \sum_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma'} d_{m''\sigma} \\ &= 2D + 2P \end{aligned}$$

The blue term can be written by considering three cases: (i) $m' = m''$, $\sigma \neq \sigma'$, (ii) $m' \neq m''$, $\sigma = \sigma'$, and (iii) $m' \neq m''$, $\sigma \neq \sigma'$. ($m' = m''$, $\sigma = \sigma'$ vanishes the electron.)

(i) $m' = m''$, $\sigma \neq \sigma'$

$$\begin{aligned} d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma'} d_{m'\sigma} &= d_{m'\sigma}^\dagger d_{m'\sigma} d_{m'\sigma'} d_{m'\sigma'} \\ &= -d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m'\sigma'} d_{m'\sigma} \\ &= -n_{m'\sigma} n_{m'\sigma'} \quad (\sim D) \end{aligned}$$

(ii) $m' \neq m''$, $\sigma = \sigma'$

$$\begin{aligned} d_{m'\sigma}^\dagger d_{m'\sigma} d_{m''\sigma} d_{m'\sigma} &= d_{m'\sigma}^\dagger d_{m''\sigma} d_{m''\sigma} d_{m'\sigma} \\ &= n_{m'\sigma} n_{m''\sigma} \quad (\sim Y) \end{aligned}$$

(iii) $m' \neq m''$, $\sigma \neq \sigma'$

$$\begin{aligned} d_{m'\sigma}^\dagger d_{m''\sigma'} d_{m''\sigma} d_{m'\sigma'} &= -d_{m'\sigma}^\dagger d_{m''\sigma'} d_{m'\sigma'} d_{m''\sigma} \\ &= d_{m'\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma'} d_{m''\sigma} \quad (\sim F) \end{aligned}$$

The summation over $\sum_{mm'} \sum_{\sigma\sigma'}$ gives

$$\sum_{mm'} \sum_{\sigma\sigma'} d_{m\sigma}^\dagger d_{m'\sigma'} d_{m''\sigma} d_{m'\sigma'} = -2D + 2Y + 2F$$

The first term of $d_{m'\sigma}^\dagger d_{m''\sigma} d_{m'\sigma}^\dagger d_{m''\sigma}$ is given by

$$\sum_{m'm''} \sum_{\sigma\sigma'} \delta_{mm'} \delta_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m''\sigma} = \sum_{m'\sigma} d_{m'\sigma}^\dagger d_{m'\sigma} = N$$

The first term of $d_{m'\sigma}^\dagger d_{m''\sigma} d_{m'\sigma}^\dagger d_{m''\sigma}$ is given by

$$\sum_{m'm''} \sum_{\sigma\sigma'} \delta_{\sigma\sigma'} d_{m'\sigma}^\dagger d_{m''\sigma} = \sum_{m'} \sum_{m'\sigma} d_{m'\sigma}^\dagger d_{m'\sigma} = 3N$$

Summing all the relations,

$$\begin{aligned} L^2 &= - \sum_{m'm''} \sum_{\sigma\sigma'} (d_{m'\sigma}^\dagger d_{m''\sigma} d_{m'\sigma}^\dagger d_{m''\sigma} - d_{m'\sigma}^\dagger d_{m''\sigma} d_{m''\sigma}^\dagger d_{m'\sigma}) \\ &= - \{ N + 2D + 2P - (3N - (-2D + 2Y + 2F)) \} \\ &= - (N + 2D + 2P - 3N - 2D + 2Y + 2F) \\ &= 2N - 2P - 2Y - 2F \end{aligned}$$

So far, one obtained:

Formula	From our notation
$\frac{N(N-1)}{2}$	$D + X + Y$
S^2	$\frac{3}{4}N - \frac{3}{2}D + \frac{1}{2}Y - \frac{1}{2}X + F$
L^2	$2N - 2P - 2Y - 2F$

Let us rewrite the H_{GE}:

$$\begin{aligned} H_{GE} &= U \sum_m n_{m\uparrow} n_{m\downarrow} + U' \sum_{m \neq m'} n_{m\uparrow} n_{m'\downarrow} + (U' - J) \sum_{m \neq m', \sigma} n_{m\sigma} n_{m'\sigma} \\ &\quad - J_x \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} + J_p \sum_{m \neq m'} d_{m\uparrow}^\dagger d_{m\downarrow} d_{m'\downarrow}^\dagger d_{m'\uparrow} \\ &= U D + U' X + (U' - J) Y - J_x F + J_p P \end{aligned}$$

Due to the rotational invariance, $J_x = J$, $J_p = U - U' - J$. Also, limiting the correlated subspace as t_{2g} manifold, $J_x = J_p$ s.t $U' = U - 2J$.

Why? Let us dig into the condition that makes these relations. First of all, one should note that it is impossible to write $\mathbb{D}$, X , Y , and F, P in terms of N, S^2 , and L^2 because the number of degrees of freedom is more larger. Here, one leaves F and P since they naturally shows how the rotational invariance breaks. One can choose other bases which lead to the same consequence. From now on, one will rewrite X, Y , and N in terms of N, S^2 , and L^2 .

$$\textcircled{a} \quad Y = \frac{1}{2}(2N - 2P - 2F - L^2) = N - P - F - \frac{1}{2}L^2$$

$$\begin{aligned} X &= Y + 2F - 3\mathbb{D} + \frac{3}{2}N - 2S^2 \\ &= N + F - P - 3\mathbb{D} + \frac{3}{2}N - \frac{1}{2}L^2 - 2S^2 \end{aligned}$$

The sum of X and Y yields

$$X + Y = 2N - 2P + \frac{3}{2}N - L^2 - 2S^2 - 3\mathbb{D} = \frac{N(N-1)}{2} - \mathbb{D}$$

$$\therefore \mathbb{D} = \frac{7}{4}N - P - \frac{1}{2}L^2 - S^2 - \frac{N(N-1)}{4}$$

Likewise,

$$\begin{aligned} \therefore X &= N + F - P - \frac{7}{4}N + 3P + \frac{3}{2}L^2 + 3S^2 + \frac{3N(N-1)}{4} \\ &\quad + \frac{3}{2}N - \frac{1}{2}L^2 - 2S^2 \\ &= -\frac{11}{4}N + L^2 + S^2 + F + 2P + \frac{3}{4}N(N-1) \end{aligned}$$

Plugging these into H_{GR} , one obtains

$$\begin{aligned} H_{GR} &= W\mathbb{D} + W'X + (W' - J)Y - J_X F + J_P P \\ &= W\left\{\frac{7}{4}N - P - \frac{1}{2}L^2 - S^2 - \frac{1}{4}N(N-1)\right\} \\ &\quad + W'\left\{-\frac{11}{4}N + L^2 + S^2 + F + 2P + \frac{3}{4}N(N-1)\right\} \end{aligned}$$

$$+ (L' - J) \left\{ N - P - F - \frac{1}{2} L^2 \right\} - J_x F + J_p P$$

$$= (-L + 2L' - L' + J + J_p) P + (L' - L' + J - J_x) F + \dots$$

The other terms are not displayed since these two terms are crucial for the symmetry argument. In order to have the rotational symmetry written as $U(1) \otimes SU(2) \otimes SO(3)$, these terms should vanish. As a matter of fact, they vanish to satisfy $[N, H_{\text{eff}}] = [S^2, H_{\text{eff}}] = [L^2, H_{\text{eff}}] = 0$.

Finally, one has the following relations:

$$J = J_x, \quad J_p = L - L' - J$$

However, the symmetry does not allow $J_p = J$. Instead, this relation is obtained by using Slater integrals for t_{2g} orbitals. Then one gets $L' = L - 2J$ which implies $J_p = J$.

What happens to other terms? If one uses $J = J_x = J_p = L - L' - J$, $L' = L - 2J$, H_{eff} is written as:

$$H_{\text{eff}} = L D + L' X + (L' - J) Y - J_x F + J_p P$$

$$\approx L \left\{ \frac{7}{4} N - \frac{1}{2} L^2 - S^2 - \frac{1}{4} N(N-1) \right\}$$

$$+ L' \left\{ -\frac{11}{4} N + L^2 + S^2 + \frac{3}{4} N(N-1) \right\} + (L' - J) \left\{ N - \frac{1}{2} L^2 \right\}$$

$$= \left\{ -\frac{L}{4} + \frac{3}{4} (L - 2J) \right\} N(N-1) + \left\{ \frac{7}{4} L - \frac{11}{4} (L - 2J) + L - 3J \right\} N$$

$$+ \left\{ -\frac{L}{2} + (L - 2J) - \frac{1}{2} (L - 3J) \right\} L^2 + (-L + L - 2J) S^2$$

$$= (L - 3J) \frac{N(N-1)}{2} + \frac{5}{2} J N - \frac{1}{2} J L^2 - 2 J S^2$$